Agentic AI

An Introduction

LLMs

- LLMs: Given a series of words (tokens) a language model, models the probability of the next word. A model for “Language Completion”
- What else?
  - The weather in Austin yesterday was …
  - John has 3 apples. He gives away 1 and then buys 4 more. He now has …
  - Please buy 300 shares of QQQ …
- Knowledge?
- Reasoning?
- Acting?

LLMs → Reasoning

- LARGE (more params, more data) models began to reason!
  - Logical Reasoning
  - Chain-of Thought Problem Solving
  - Common Sense Inference
  - Mathematical Reasoning
- Why does this happen?
- It is imitated/simulated reasoning, not exactly the same as human reasoning
  - don’t have internal models of logic or facts
  - reasoning can be inconsistent or brittle
  - don’t “understand” in a human sense; they pattern-match effectively

Actions

- Write an Email Vs Send an Email

Reasoning is central

- Tool use and delegation
    - Agents often use tools (e.g., calculators, APIs) or even other agents.
    - It must reason about when and how to use those tools effectively.
- Goal-directed planning
    - “If I want X, then I must do Y and Z first.” Requires multi-step reasoning, not just reacting.
- Decision-making
    - Choosing between options requires evaluating outcomes, costs, risks — all forms of reasoning.
- Self-reflection and correction
    - An agent needs to reason about its own performance and revise plans if they’re not working.

Agents: Wrappers around LLMs

- Core differentiators
  - Tools : take *actions* in the world (query a DB, send email, call APIs)
  - Looping / Autonomy : keeps reasoning + acting until the goal is complete
- Additional capabilities
  - Knowledge (RAG, KB etc)
  - Memory (beyond context window, short/long term)
  - Goals (multi-step planning)
  - Control & Guardrails (safety, limits, permissions)
  - Planning / Reasoning processes (ReAct, Plan-and-Execute)
  - Environment / Context awareness (time, state, user preferences)
  - Collaboration abilities (multi-agent ecosystems, delegation)
  - Personality / Role (shaping communication style and behavior)